

Software Engineer | Project Assignment

The **Personal Calorie Tracker** is a full-stack application designed to help users monitor, manage, and understand their daily nutritional intake. Users can log meals across breakfast, lunch, and dinner, set personalized health goals, and visualize macro and micronutrient trends over time.

Requirements

- **Goal Setting:** Ability to set and manage personal health goals (e.g., daily calorie target, protein/carb/fat targets, weight goal) through the web app.
- **Meal Entry:** Ability to create food entries grouped by meal type (Breakfast, Lunch, Dinner, Snacks) with fields for food item name, quantity, and nutritional values (calories, macros, micros).
- **Time-Range Listing:** List all food entries in a specified time range through the web app, filterable by date and meal type.
- **Nutrition Reports & Graphs:** Ability to display visual reports including: weekly calorie intake trend, macronutrient breakdown (protein, carbs, fat) by day/week, micronutrient summary (vitamins, minerals), and goal vs. actual comparison charts.
- **AI-Powered Calorie Extraction:** Ability to upload a photo (product nutrition label or a plate of food) and automatically extract and pre-fill calorie and nutritional information using AI image analysis.

Data Model & Programming Choices

- Use the data model that best suits the above requirements.
- APIs must be separate from frontend code. The frontend web app will communicate with the backend exclusively through APIs.
- Persist all food entries, goals, and user data in a database.
- Support pagination in all list APIs.

Bonus Features (Extra Credit)

- **Conversational Chat Interface:** Build a chat interface powered by an LLM that allows users to perform all app actions through natural language — logging meals, checking goals, asking nutritional questions, and getting weekly summaries — without touching traditional UI controls.

- **Multi-User Support:** Support multiple independent users who can sign up, log in, and maintain their own private data.
- **Bulk Import via PDF:** Support upload of a food diary or nutrition history exported as a PDF (tabular format) and automatically parse and import the entries.

Code Quality Guidelines

The quality of the code you submit will play a significant role in our evaluation process. Here are the specific guidelines we will be looking for:

1. **Clean Code:** Write clear, concise code that is easy to read and understand. Use meaningful variable and function names. Avoid overly complex or convoluted structures.
2. **Modularity:** Break your code into logical modules or functions, promoting reusability and maintainability. Avoid monolithic code.
3. **Documentation:** Provide a README file with clear instructions on how to set up and run the application, along with any assumptions made.
4. **Error Handling:** Implement proper error handling and input validation to ensure robustness and a good user experience.
5. **Comments (Optional):** Include appropriate inline comments to explain complex logic or important architectural decisions.

Please allocate sufficient time and effort to ensure your submission reflects your best coding skills and engineering judgment. The quality of your work will be a crucial factor in moving forward.